

Data Quality and Its Dimensions

for a Split-Federated Learning Setting

A Targeted Literature Review

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System Context

Setting: Split-federated learning (SplitFed) applied to household smart-meter load forecasting at hourly resolution.

Conditions: Non-stationary environment, endemic concept drift, heterogeneous clients, resource-constrained edge devices, strict privacy preservation (no raw data sharing).

Selected DQ Dimensions: Accuracy • Completeness • Temporal Consistency • Consistency.

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1 Introduction

Data quality (DQ) has been recognised as a foundational prerequisite for reliable data-driven systems since at least the mid-1990s. The landmark empirical study by [Wang and Strong \(1996\)](#) identified fifteen dimensions across which data consumers and producers assess quality. [Pipino et al. \(2002\)](#) subsequently operationalised many of these dimensions as objective, ratio-based metrics, and [Batini et al. \(2009\)](#) provided a comprehensive methodological survey covering more than fifty DQ assessment and improvement approaches.

In a classical centralised setting, DQ assessment is tractable: a data steward can access the full dataset and compute global quality metrics. In federated and split-federated settings, this assumption collapses fundamentally. Raw data remain on local devices; only model updates—gradients or intermediate activations—are transmitted to a coordinating server. This privacy-preserving constraint means that global DQ metrics cannot be computed directly and must either be estimated locally and securely aggregated, or inferred from model behaviour ([Kairouz et al., 2021](#)).

For household smart-meter forecasting, the challenge is compounded by the inherently non-stationary character of energy consumption. Occupancy patterns change seasonally, appliances are replaced, tariff structures evolve, and concept drift is endemic in long-running deployments ([Gama et al., 2014](#)). On heterogeneous, resource-constrained edge devices, data collection may be interrupted, device clocks may drift, and local preprocessing inconsistencies may be invisible to the central server.

This review focuses on the four DQ dimensions that are both theoretically grounded in the classical DQ literature and directly actionable within a privacy-preserving SplitFed, edge-device system: **Accuracy**, **Completeness**, **Temporal Consistency**, and **Consistency**. For each dimension the review traces (i) the conceptual definition, (ii) how different papers compute a quantitative score, and (iii) the specific adaptations required for the SplitFed context. The section on Temporal Consistency additionally reconciles the diverse terminology used across database, machine learning, and MLOps communities, and motivates and references four concrete subchecks essential for hourly-resolution forecasting under concept drift. A final section justifies the selection of these four dimensions over alternatives.

2 Conceptual Foundations of Data Quality Dimensions

2.1 The Wang and Strong (1996) Taxonomy

Wang and Strong (1996) conducted an empirical study with data consumers and producers, identifying 179 DQ attributes that were condensed into fifteen dimensions grouped into four higher-level categories:

- *Intrinsic* DQ: accuracy, objectivity, believability, reputation.
- *Contextual* DQ: relevancy, value-added, timeliness, completeness, appropriate amount.
- *Representational* DQ: interpretability, ease of understanding, concise and consistent representation.
- *Accessibility* DQ: accessibility, access security.

The taxonomy serves as a starting point for dimension selection. Its primary value in the SplitFed context is in distinguishing *intrinsic* dimensions (computable locally from the data itself without external reference) from contextual and representational dimensions (which often require external judgment or schema-level inspection).

2.2 Pipino, Lee and Wang (2002): Operationalisation

Pipino et al. (2002) is the foundational reference for *quantitative* DQ scoring. They propose two classes of metrics applicable to any dimension. *Simple ratio* metrics take the form:

$$q = \frac{d}{D}, \quad (1)$$

where d is the count of data units satisfying the quality criterion and D is the total count of data units. *Weighted ratio* metrics aggregate over multiple sub-criteria:

$$q_{\text{weighted}} = \sum_{k=1}^K w_k \cdot \frac{d_k}{D_k}, \quad \sum_{k=1}^K w_k = 1. \quad (2)$$

Both metrics produce scores in $[0, 1]$, which is a critical property for the SplitFed context: local scores can be safely communicated to the server and aggregated without revealing any information about the underlying data values.

2.3 Batini et al. (2009): Methodological Survey

Batini et al. (2009) survey more than fifty DQ assessment and improvement methodologies. They distinguish *data-driven* methods (computed from the data itself using statistical tests or semantic constraints) from *process-driven* methods (assessed by auditing the data production process). For the SplitFed context, only data-driven methods executed locally on-device are feasible, and the survey provides the primary methodological grounding for the scoring formulas reviewed below.

3 Dimension 1: Accuracy

3.1 Definition

Accuracy is the degree to which data values correctly represent the real-world entities they are intended to describe (Wang and Strong, 1996). Batini et al. (2009) further decompose accuracy into *syntactic accuracy*—conformance to a domain format or value range—and *semantic accuracy*—agreement with the actual ground truth. For smart-meter readings, syntactic accuracy corresponds to range validity (non-negative power consumption, plausible peak demand bounds), while semantic accuracy would require comparison against reference meters—typically infeasible in a privacy-preserving federated setting.

3.2 Scoring Methods in the Literature

Pipino, Lee and Wang (2002). The simplest and most widely cited formulation is the ratio of correct values to total values:

$$\text{Acc}_{\text{PLW}} = \frac{|\{x_i : x_i \text{ is correct}\}|}{N}. \quad (3)$$

Pipino et al. (2002) acknowledge that “correct” must be defined contextually for each dataset. For numeric sensor readings, this is operationalised as membership in a physically plausible domain $[\ell, u]$, so that:

$$\text{Acc}_{\text{PLW}} = \frac{|\{x_i : \ell \leq x_i \leq u\}|}{N}. \quad (4)$$

Batini et al. (2009). Batini et al. (2009) express accuracy as the complement of the error rate:

$$\text{Acc}_{\text{BAT}} = 1 - \frac{n_e}{N}, \quad (5)$$

where n_e is the count of detected erroneous values—out-of-range, physically impossible, or contradicted by a redundant source. This is equivalent to Equation (3) when valid and erroneous values are mutually exclusive. Batini et al. distinguish syntactic errors (identifiable by range or format checks) from semantic errors (requiring an external oracle), and note that only syntactic accuracy can be assessed automatically.

Outlier-based accuracy for time series. In the time-series domain, accuracy is frequently operationalised through anomaly detection against a local predictive model. A reading x_t is flagged as inaccurate if it deviates from a local one-step-ahead forecast $\hat{x}_t$ by more than a threshold δ :

$$\text{Acc}_{\text{TS}} = 1 - \frac{1}{T} \sum_{t=1}^T \mathbf{1}[|x_t - \hat{x}_t| > \delta]. \quad (6)$$

This approach is particularly valuable for edge devices because the local model $\hat{x}_t$ (e.g., a simple AR or exponential smoothing model) can be estimated from recent history without sharing raw data. Dasu and Johnson (2003) describe local-model deviation as a standard component of exploratory data analysis for sensor streams.

Weighted multi-rule accuracy for energy data. Domain-specific accuracy rules for smart-meter data include: consumption values must be non-negative; hourly totals must be consistent with sub-interval readings; reactive power must not exceed apparent power. A weighted multi-rule accuracy score following the framework of Pipino et al. (2002) is:

$$\text{Acc}_{\text{rule}} = \frac{\sum_{r \in \mathcal{R}} w_r \cdot \frac{|\{x_t : x_t \text{ satisfies rule } r\}|}{T}}{\sum_{r \in \mathcal{R}} w_r}, \quad (7)$$

where $\mathcal{R}$ is the set of domain rules and w_r is the criticality weight of rule r . Haben et al. (2021) note that energy data frequently contain anomalous readings introduced by meter tampering, communication errors, or clock misconfiguration, making rule-based

checks an important first-pass quality screen.

Table 1: Summary of accuracy scoring methods reviewed.

Source / Method	Formula	Quality criterion
Pipino et al. (2002)	d/D	Count of values meeting domain criteria
Batini et al. (2009)	$1 - n_e/N$	Complement of detected error rate
Time-series anomaly detection (Dasu and Johnson, 2003)	$1 - \frac{1}{T} \sum_t \mathbf{1}[x_t - \hat{x}_t > \delta]$	Local AR/ETS model deviation
Weighted rule-based (Pipino et al., 2002)	$\sum_r w_r \cdot d_r/D_r$	Physical and domain constraint satisfaction

3.3 Accuracy in the SplitFed Context

Each client k computes a local accuracy score Acc_k using either the range-validity check or the local-model deviation approach. The global accuracy estimate aggregates these scores weighted by the number of local records:

$$\widehat{\text{Acc}}_{\text{global}} = \sum_{k=1}^K \frac{N_k}{\sum_j N_j} \cdot \text{Acc}_k. \quad (8)$$

Because raw data are never shared, semantic accuracy against an external ground truth cannot be verified globally. This is an inherent and acknowledged limitation of privacy-preserving federated systems (Kairouz et al., 2021).

4 Dimension 2: Completeness

4.1 Definition

Completeness measures the extent to which all expected data values are present (Wang and Strong, 1996). Batini et al. (2009) distinguish four levels: *schema completeness* (all required attributes exist in the schema), *column completeness* (no null values in an attribute column), *population completeness* (all real-world entities are represented in the dataset), and *record completeness* (all attributes of each record are populated). For smart-meter time series, these four levels reduce in practice to *temporal completeness*:

the fraction of expected hourly readings that are actually present and non-null.

4.2 Scoring Methods in the Literature

Pipino, Lee and Wang (2002). The simple ratio framework gives:

$$\text{Comp}_{\text{PLW}} = \frac{v}{D}, \quad (9)$$

where v is the count of non-null values and D is the count of expected values. A fully populated dataset scores 1.0; each missing value reduces the score by $1/D$.

Batini and Scannapieco (2016). Batini and Scannapieco (2016) formalise attribute-level and tuple-level completeness as distinct metrics:

$$\text{Comp}_{\text{attr}} = 1 - \frac{|\text{NULL values in attribute}|}{|R|}, \quad \text{Comp}_{\text{tuple}} = \frac{|\{r \in R : \nexists a_i \text{ s.t. } r.a_i = \text{NULL}\}|}{|R|}, \quad (10)$$

where R is the set of records and a_i iterates over attributes. Tuple completeness is a stricter criterion, requiring that a record be fully populated across all attributes to count as complete.

Temporal completeness for regularly sampled time series. For time series sampled at a fixed interval Δt , the expected number of readings over a window $[t_{\text{start}}, t_{\text{end}}]$ is $T_{\text{exp}} = \lfloor (t_{\text{end}} - t_{\text{start}}) / \Delta t \rfloor + 1$. Temporal completeness is then:

$$\text{Comp}_{\text{temporal}} = 1 - \frac{n_{\text{missing}}}{T_{\text{exp}}}, \quad (11)$$

where n_{missing} counts gaps in the timestamp sequence. Haben et al. (2021) report that missingness rates in real-world smart-meter datasets commonly range from 1% to 15%, and that *block missingness*—contiguous gaps of multiple consecutive readings—is substantially more damaging to forecasting quality than scattered missingness at the same overall rate, because it disrupts the autocorrelation structure that forecasting models rely on.

Gap-severity weighted completeness. To penalise block missingness more than scattered missingness, a severity-weighted score replaces the raw gap count with a

superlinear penalty. If gap g spans ℓ_g consecutive missing readings:

$$\text{Comp}_{\text{gap}} = 1 - \frac{\sum \phi(\ell_g)}{T_{\text{exp}}}, \quad (12)$$

where $\phi(\ell) = \ell^\alpha$ with $\alpha > 1$ assigns disproportionately higher penalty to longer gaps. Setting $\alpha = 1$ recovers Equation (11).

Table 2: Summary of completeness scoring methods reviewed.

Source / Method	Formula	Quality criterion
Pipino et al. (2002)	v/D	Non-null to expected values ratio
Batini and Scannapieco (2016)	$1 - \text{NULL} / R $	Attribute- or tuple-level null fraction
Temporal completeness (Haben et al., 2021)	$1 - n_{\text{miss}}/T_{\text{exp}}$	Fraction of expected timestamps present
Gap-severity weighted	$1 - \sum_g \ell_g^\alpha / T_{\text{exp}}$	Superlinear penalty for block gaps

4.3 Completeness in the SplitFed Context

Completeness is the DQ dimension most naturally computed locally: each device knows its expected sampling interval and can count missing timestamps without any external reference. Crucially, local completeness scores reveal nothing about the underlying energy values and can be safely reported to the server. The server can use these scores to weight client contributions during federated aggregation, assigning lower influence to clients whose training windows contain substantial missingness (Kairouz et al., 2021).

5 Dimension 3: Temporal Consistency

5.1 Definition and Alternative Terminology

Temporal consistency is the most multifaceted DQ dimension for time-series forecasting systems operating under concept drift. It captures the idea that data should be *current*, *appropriately aligned in time*, and *drawn from a distribution that remains*

consistent with the model’s assumptions. The literature employs a proliferation of overlapping terms for closely related concepts, which must be reconciled before a coherent scoring framework can be constructed:

- **Currency** (Wang and Strong, 1996): the degree to which data are sufficiently up-to-date for the task at hand.
- **Timeliness** (Batini et al., 2009): a composite of currency and volatility; data are timely when they are current relative to how rapidly the underlying real-world state changes.
- **Freshness** (Aggarwal, 2007): the inverse of data age relative to the moment of use; older data are less fresh.
- **Staleness**: the complement of freshness; the degree to which data have become outdated beyond an acceptable threshold.
- **Temporal validity** (Dasu and Johnson, 2003): whether each record’s timestamp falls within the valid operational window.
- **Recency** (Sidi et al., 2012): the proportion of records whose timestamps fall within a defined recency window relative to the current time.
- **Concept drift** (Gama et al., 2014): a change in the joint distribution $P(X, Y)$ over time, which can manifest as covariate shift $P(X)$, label shift $P(Y)$, or posterior shift $P(Y | X)$.
- **Dataset shift** (Rabanser et al., 2019): a broad term encompassing any change in the statistical distribution of data between training and deployment time.
- **Train–serve skew** (Sculley et al., 2015): a discrepancy between the distribution of features encountered during model training and those encountered at inference time, arising from pipeline differences, delayed feature updates, or genuine distributional change.

These terms are used inconsistently across communities. The database DQ community emphasises currency and timeliness; the machine learning community emphasises concept drift and dataset shift; the MLOps community emphasises train–serve skew. For a SplitFed forecasting system under non-stationary conditions, all three framings are simultaneously relevant and must be unified within a single operational dimension.

5.2 Subcheck 1: Staleness of the Training Window

Batini et al. (2009) propose an age-based formulation for currency. Given a record or window w last updated at time t_{update} , its age is:

$$\text{Age}(w) = t_{\text{now}} - t_{\text{update}}, \quad (13)$$

and the normalised currency score is:

$$\text{Curr}(w) = \max\left(0, 1 - \frac{\text{Age}(w)}{\tau}\right), \quad (14)$$

where τ is the acceptable staleness threshold—for day-ahead smart-meter forecasting, a common choice is $\tau = 48$ hours. A training window is flagged as stale when the gap between its most recent observation and the current timestamp exceeds τ , producing $\text{Curr} = 0$. Pipino et al. (2002) note that currency, like all DQ dimensions, should be expressed as a normalised ratio in $[0, 1]$, making Equation (14) a direct instantiation of their general framework.

5.3 Subcheck 2: Distributional Drift of Inputs and Target

Population Stability Index (PSI). The most widely deployed operational metric for distributional drift is the *Population Stability Index*, originating in the credit-risk literature and subsequently adopted throughout the MLOps community for monitoring feature and target distributions. Given B equal-width or equal-frequency bins:

$$\text{PSI} = \sum_{i=1}^B (A_i - E_i) \ln \frac{A_i}{E_i}, \quad (15)$$

where A_i is the fraction of current observations falling in bin i and E_i is the corresponding fraction in the reference (training) distribution. Standard thresholds are: $\text{PSI} < 0.10$ indicates negligible shift; $0.10 \leq \text{PSI} < 0.25$ indicates moderate shift warranting investigation; $\text{PSI} \geq 0.25$ indicates significant shift requiring model retraining. A normalised DQ score is:

$$\text{DriftScore}_{\text{PSI}} = 1 - \min\left(1, \frac{\text{PSI}}{0.25}\right). \quad (16)$$

Kullback–Leibler Divergence. A theoretically principled alternative is the Kullback–Leibler (KL) divergence:

$$D_{\text{KL}}(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}, \quad (17)$$

where Q is the reference distribution and P is the current distribution. Note that PSI is symmetric in A and E and equals $D_{\text{KL}}(A\|E) + D_{\text{KL}}(E\|A)$.

Maximum Mean Discrepancy and classifier-based tests. Rabanser et al. (2019) provide an empirical benchmark of distributional shift detection methods including PSI, KL divergence, Maximum Mean Discrepancy (MMD), and classifier-based two-sample tests. They find that PSI and MMD are robust across a wide range of shift magnitudes for tabular data. For multivariate inputs such as a vector of lagged energy readings combined with exogenous features, MMD is often preferred because it naturally handles multivariate distributions without requiring binning.

Adaptive change detectors for streaming data. Gama et al. (2014) review statistical process control (SPC) methods, the Page–Hinkley Test (PHT), and the ADWIN algorithm (Bifet and Gavaldà, 2007) for detecting concept drift in streaming settings. ADWIN maintains an adaptive sliding window and flags drift when the empirical mean of a monitored statistic changes beyond a threshold, providing a principled way to detect both sudden and gradual drift in local edge-device data streams.

5.4 Subcheck 3: Calendar Alignment of Hourly/Weekly Seasonality

Hourly smart-meter time series exhibit strong and predictable weekly and daily periodicity. Hong et al. (2016) emphasise that energy forecasting models must cover the full spectrum of load conditions, including weekday/weekend differences and seasonal variation; a training window that systematically omits certain hours or calendar days will produce biased forecasts for those periods.

A calendar coverage score checks that the training window contains a sufficient and representative sample of all hour-of-day $\times$ day-of-week combinations:

$$\text{Cal} = \frac{1}{H \cdot D_w} \sum_{h=0}^{H-1} \sum_{d=1}^{D_w} \mathbf{1}[n_{h,d} \geq n_{\min}], \quad (18)$$

where $H = 24$ is the number of hours in a day, $D_w = 7$ is the number of days in a week, $n_{h,d}$ is the count of observations for hour h and weekday d in the training

window, and $n_{\min}$ is the minimum required count (typically corresponding to two to four weeks of coverage). A score of 1.0 indicates that all $H \times D_w = 168$ hour-of-day $\times$ day-of-week cells are adequately represented.

5.5 Subcheck 4: Train–Serve Mismatch

[Sculley et al. \(2015\)](#) introduce train–serve skew as a source of hidden technical debt in production ML pipelines: the distribution of features used during training may differ from the distribution encountered at inference time due to pipeline differences, delayed feature computation, or genuine distributional shift. In the SplitFed context, train–serve mismatch has an additional and specific meaning: the server maintains a global model trained on an aggregated representation of all clients, while each client serves predictions against its own local distribution, which may have drifted from the global reference since the last federation round.

The train–serve mismatch score is operationalised as the PSI between the local client’s current feature distribution and the reference distribution active at the time of the last global model update:

$$\text{Mismatch}_k = \text{PSI}(P_k^{\text{current}}, P^{\text{ref}}), \quad (19)$$

where P^{ref} is the reference distribution encoded in the server’s global model. In a privacy-preserving deployment, P^{ref} can be approximated by summary statistics (per-feature mean and variance) maintained at the server and broadcast to clients at the start of each federation round. High Mismatch_k values signal that client k has drifted significantly from the global model’s assumptions, warranting local fine-tuning, a personalised federation round, or flagging for data relabelling.

5.6 Composite Temporal Consistency Score

The four subchecks are combined into a single composite score:

$$\text{TC} = w_1 \cdot \text{Curr} + w_2 \cdot \text{DriftScore} + w_3 \cdot \text{Cal} + w_4 \cdot (1 - \text{Mismatch}_{\text{norm}}), \quad w_1 + w_2 + w_3 + w_4 = 1, \quad (20)$$

where $\text{Mismatch}_{\text{norm}} = \min(1, \text{Mismatch}_k / 0.25)$. In a non-stationary smart-meter deployment, distributional drift and train–serve mismatch are likely to be the most critical subchecks and may warrant higher weights relative to staleness and calendar

alignment.

Table 3: Summary of temporal consistency subchecks, associated terminology, and references.

Subcheck	Alternative terms	Key references	Metric
Staleness of training window	Currency, Timeliness, Freshness, Recency	Batini et al. (2009); Wang and Strong (1996)	Age-based ratio: $\max(0, 1 - \text{Age}/\tau)$
Distributional drift	Dataset shift, Concept drift, Covariate shift	Rabanser et al. (2019); Gama et al. (2014); Bifet and Gavaldà (2007)	PSI, KL divergence, MMD, ADWIN
Calendar alignment	Seasonal coverage, Load condition coverage	Hong et al. (2016)	Hour $\times$ weekday cell coverage ratio
Train-serve mismatch	Train-serve skew, Serving shift	Sculley et al. (2015)	PSI between client current and server reference distributions

6 Dimension 4: Consistency

6.1 Definition

Consistency measures the degree to which data satisfy a set of semantic rules or integrity constraints (ICs) and do not contradict one another (Batini et al., 2009). It differs from accuracy in that consistency is assessed *internally*—data values must agree with each other and with defined rules—without requiring access to an external ground truth. In the SplitFed smart-meter context, consistency encompasses three levels: (i) *intra-device consistency*: readings from the same meter at successive timestamps must satisfy physical rate-of-change constraints; (ii) *inter-feature consistency*: if sub-meter readings are available, their sum must equal the total meter reading; (iii) *cross-device consistency*: readings from meters within the same building or substation must obey aggregate energy balance constraints, though this third level requires sharing at least aggregated statistics.

6.2 Scoring Methods in the Literature

Batini et al. (2009). The standard formulation is the ratio of records satisfying all applicable ICs:

$$\text{Cons}_{\text{BAT}} = \frac{|\{r \in R : r \models \mathcal{I}\}|}{|R|}, \quad (21)$$

where $\mathcal{I}$ is the set of integrity constraints and $r \models \mathcal{I}$ denotes that record r satisfies every constraint in $\mathcal{I}$. Each constraint violation is counted once regardless of severity.

Weighted constraint satisfaction (Pipino et al., 2002). Pipino et al. (2002) extend this to a weighted formulation that assigns higher weight to constraints of greater criticality:

$$\text{Cons}_{\text{weighted}} = \sum_{c \in \mathcal{I}} w_c \cdot \frac{|\{r \in R : r \models c\}|}{|R|}, \quad \sum_{c \in \mathcal{I}} w_c = 1. \quad (22)$$

Physical plausibility constraints (e.g., non-negative instantaneous demand) receive higher weight than soft business rules (e.g., typical peak demand thresholds for a household meter class).

Rate-of-change consistency. For time series, a standard consistency check verifies that successive differences do not exceed a physically plausible bound Δ_{max} :

$$\text{Cons}_{\text{ROC}} = 1 - \frac{1}{T-1} \sum_{t=2}^T \mathbf{1}[|x_t - x_{t-1}| > \Delta_{\text{max}}]. \quad (23)$$

Dasu and Johnson (2003) describe rate-of-change checks as a standard component of exploratory data analysis for sensor streams, noting that they detect both sensor faults (sudden large spikes) and communication errors (readings that are stale replicas of an earlier value, producing a zero difference for an extended period).

Cross-feature consistency. When sub-meter readings are available, cross-feature consistency verifies the energy balance:

$$\text{Cons}_{\text{cross}} = 1 - \frac{1}{T} \sum_{t=1}^T \mathbf{1}\left[\left|\sum_{j=1}^J x_t^{(j)} - x_t^{\text{total}}\right| > \varepsilon\right], \quad (24)$$

where $x_t^{(j)}$ denotes sub-meter j at time t and ε is a tolerance for rounding.

Table 4: Summary of consistency scoring methods reviewed.

Source / Method	Formula	Quality criterion
Batini et al. (2009)	$ \{r : r \models \mathcal{I}\} / R $	Fraction of records satisfying all ICs
Pipino et al. (2002)	$\sum_c w_c \cdot d_c / R $	Weighted IC satisfaction ratio
Rate-of-change check (Dasu and Johnson, 2003)	$1 - \frac{1}{T-1} \sum_t \mathbf{1}[\Delta x_t > \Delta_{\max}]$	Successive difference within physical bound
Cross-feature balance	$1 - \frac{1}{T} \sum_t \mathbf{1}[\sum_j x_t^{(j)} - x_t^{\text{total}} > \varepsilon]$	Sub-meter aggregate equals total meter

6.3 Consistency in the SplitFed Context

Intra-device and inter-feature consistency can be computed locally without sharing raw data. Cross-device consistency requires aggregated statistics, but in the SplitFed architecture this can be approximated via the intermediate activations already transmitted for training purposes. Thapa et al. (2022) note that the cut-layer activations transmitted from client to server in SplitFed carry a compressed representation of the local input, which the server can use to monitor distributional consistency across clients without accessing raw readings.

7 Justification for Dimension Selection in the SplitFed Context

7.1 Why These Four Dimensions Are Appropriate

The selection of Accuracy, Completeness, Temporal Consistency, and Consistency is grounded jointly in the foundational DQ taxonomy of Wang and Strong (1996) and in the specific operational requirements of the SplitFed smart-meter problem.

Accuracy is the baseline requirement for any forecasting system. Inaccurate sensor readings corrupt both the target variable and the feature inputs, directly degrading model performance. In a federated setting, accuracy errors from a single client propagate into the global model and adversely affect predictions for all other clients (Kairouz

et al., 2021). The dimension is locally computable, score-based, and does not require raw data to be shared.

Completeness is critical because edge smart meters frequently suffer from communication interruptions, power outages, and network congestion, producing missing readings. Haben et al. (2021) document that missingness rates of 1%–15% are common in real smart-meter deployments and that block missingness is disproportionately damaging to load forecasting accuracy relative to its absolute magnitude. Temporal completeness scores are safe to report to the server and can directly inform aggregation weights.

Temporal Consistency is the dimension most specific to the non-stationary, concept-drift-prone environment described in the problem statement. No other dimension captures staleness, distributional drift, calendar coverage, or train–serve mismatch. Gama et al. (2014) demonstrate that undetected concept drift is the primary cause of model degradation in streaming and online learning settings—a finding directly transferable to continual federated learning. Sculley et al. (2015) identify train–serve skew as one of the most prevalent and operationally damaging sources of hidden technical debt in production ML pipelines.

Consistency captures structural relationships within and across data streams that the other three dimensions do not address. A reading can be accurate (within physical bounds), complete (not missing), and temporally current yet still violate physical rate-of-change constraints or contradict simultaneous sub-meter readings. Dasu and Johnson (2003) establish rate-of-change and cross-feature checks as standard practice for sensor data quality, and Thapa et al. (2022) show that the SplitFed architecture provides a natural mechanism for cross-client consistency monitoring via shared intermediate representations.

7.2 Dimensions Excluded and Their Rationale

Several DQ dimensions identified by Wang and Strong (1996) are weak or redundant for this setting:

Believability and *objectivity* are subjective, survey-based dimensions that cannot be operationalised for automated, on-device DQ assessment. *Accessibility* is a systems-engineering concern pertaining to storage and retrieval infrastructure, not a statistical property of the data itself. *Relevancy* and *value-added* depend on task context and user judgement and are therefore unsuitable for automated scoring in a federated pipeline. *Representational* dimensions (interpretability, ease of understanding, con-

ciseness) are schema-level properties that do not vary at the window or record level and thus provide no information for per-window DQ scoring. *Timeliness* as defined by Wang and Strong (1996)—and Batini et al. (2009)—is fully subsumed within Temporal Consistency; treating them as separate dimensions would introduce redundancy without additional discriminative power.

8 Conclusion

This review has systematically examined how Accuracy, Completeness, Temporal Consistency, and Consistency are defined, measured, and scored across the data quality literature, with specific attention to their relevance and adaptability for split-federated learning applied to smart-meter hourly forecasting.

The principal contributions of the review are threefold. First, it traces the scoring formulas for each dimension across multiple foundational papers—Wang and Strong (1996), Pipino et al. (2002), Batini et al. (2009); Batini and Scannapieco (2016), and Dasu and Johnson (2003)—demonstrating that all four dimensions converge on ratio-based scores in $[0, 1]$ that are computable locally on edge devices without sharing raw data. Second, it reconciles the fragmented terminology around Temporal Consistency—including currency, timeliness, freshness, staleness, temporal validity, recency, concept drift, dataset shift, and train-serve skew—into a single operationally coherent dimension with four measurable subchecks, each supported by dedicated references. Third, it justifies the selection of these four dimensions over alternatives available in the literature, demonstrating that the excluded dimensions are either redundant, subjective, or not actionable in a privacy-preserving SplitFed deployment.

Taken together, the four-dimension framework provides the foundation for a local DQ scoring module that each client executes independently, whose scalar outputs can be safely reported to and aggregated at the server, and whose scores can inform adaptive aggregation weighting, selective client participation, and drift-triggered retraining within the federated learning pipeline.

References

- Aggarwal, C. C. (2007). *Data Streams: Models and Algorithms*. Springer, New York.
- Batini, C., Cappiello, C., Francalanci, C., and Mussi, A. (2009). Methodologies for data quality assessment and improvement. *ACM Computing Surveys*, 41(3):16:1–16:52.
- Batini, C. and Scannapieco, M. (2016). *Data and Information Quality: Dimensions, Principles and Techniques*. Springer, Cham.
- Bifet, A. and Gavaldà, R. (2007). Learning from time-changing data with adaptive windowing. In *Proceedings of the 7th SIAM International Conference on Data Mining (SDM)*, pages 443–448. SIAM.
- Dasu, T. and Johnson, T. (2003). *Exploratory Data Mining and Data Cleaning*. Wiley-Interscience, Hoboken, NJ.
- Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., and Bouchachia, A. (2014). A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4):44:1–44:37.
- Haben, S., Arora, S., Giasemidis, G., Voss, M., and Greetham, D. V. (2021). Review of low voltage load forecasting: Methods, applications, and recommendations. *Applied Energy*, 304:117798.
- Hong, T., Pinson, P., Fan, S., Zareipour, H., Troccoli, A., and Hyndman, R. J. (2016). Probabilistic energy forecasting: Global energy forecasting competition 2014 and beyond. *International Journal of Forecasting*, 32(3):896–913.
- Kairouz, P., McMahan, H. B., Avent, B., Bellet, A., Bennis, M., et al. (2021). Advances and open problems in federated learning. *Foundations and Trends in Machine Learning*, 14(1–2):1–210.
- Pipino, L. L., Lee, Y. W., and Wang, R. Y. (2002). Data quality assessment. *Communications of the ACM*, 45(4):211–218.
- Rabanser, S., Günnemann, S., and Lipton, Z. C. (2019). Failing loudly: An empirical study of methods for detecting dataset shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32. Curran Associates.

- Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., and Dennison, D. (2015). Hidden technical debt in machine learning systems. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 28. Curran Associates.
- Sidi, F., Shariat Panahy, P. H., Affendey, L. S., Jabar, M. A., Ibrahim, H., and Mustapha, A. (2012). Data quality: A survey of data quality dimensions. In *2012 International Conference on Information Retrieval & Knowledge Management*, pages 300–304. IEEE.
- Thapa, C., Arachchige, P. C. M., Camtepe, S., and Sun, L. (2022). SplitFed: When federated learning meets split learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 8485–8493.
- Wang, R. Y. and Strong, D. M. (1996). Beyond accuracy: What data quality means to data consumers. *Journal of Management Information Systems*, 12(4):5–33.